Iter1.4 Validation

<G3T2>

Validation Report

Our team conducted user validation with 6 participants from our target demographic to evaluate whether our application effectively addresses two key pain points:

- Difficulty starting tasks due to high activation energy
- Internal bargaining ("I'll do it later") without real consequences

The validation focused specifically on two core actions:

- Creating a pact
- Submitting evidence for a completed pact

We conducted a two-day validation study, which allowed us to assess short-term behavior instead of relying purely on hypothetical feedback.

Methodology

We had a one-on-one interview with 6 university students (ages 19–25), and had them test out our clickable prototype in Figma.

All participants:

- Self-identified as frequent procrastinators
- Currently had one task due within 48 hours, but wants to complete it today
- Reported experiencing guilt after delaying tasks

Day 1 – Pact Creation Session (20 minutes)

Procedure:

1. Participants created a pact within the prototype.
2. They selected a meaningful consequence.
3. They answered the interviewer's questions.
4. To simulate in-app reminders, participants were messaged at:
    - 6 hours remaining
    - 2 hours remaining
    - 1 hour remaining

Participants were told to return the next day for follow-up.

Day 2 – Evidence Submission (20 minutes)

If participants completed their task, they would submit evidence through the prototype.
If participants did not complete their task, they would simulate the submitting of evidence.

Participants were asked a few questions afterwards.

Validation Limitations

Due to the short 24-hour timeframe, this study evaluates short-term behavioral influence rather than long-term habit formation. Further longitudinal testing would be required to measure sustained behavior change.

Key Findings

A. Consequence Selection Behaviour

All 6 participants selected relatively simple or low-severity consequences (e.g., small forfeits, minor inconveniences).

None selected highly uncomfortable or financially meaningful consequences.

Some participants reported that they “chose something I wouldn’t mind doing.”

Interpretation:
Participants naturally avoid setting consequences that are too painful. This suggests that users may optimize for psychological comfort rather than effectiveness when left fully in control.

This finding challenges our assumption that users will self-impose sufficiently strong consequences to change behavior.

Recommendation:

- Allow random or system-suggested consequences that users will only discover if they fail the task.
- Have the “validator” set the consequence (usually a friend would set a “harder” consequence)

B. Task Completion Outcome

Out of 6 participants:

- 4 completed their task within the agreed timeframe
- 2 did not complete their task

During Day 2 reflection:

- None of the participants stated that the consequence was an important factor in completing their tasks
- Many participants admitted they chose something easy because they did not want real discomfort

Interpretation:
While a majority completed their tasks, the perceived pressure from consequences was weaker than expected. The low-severity consequences may have reduced the mechanism's effectiveness.

This suggests that the strength of behavioral influence is dependent on consequence severity.

Recommendation:

- Introduce stronger, socially or personally meaningful consequences to increase perceived accountability.

C. Pact Creation Flow Increases Activation Energy

Participants were required to complete more than 10 interactive steps to create a pact.

- 4/6 participants commented that the process felt "long"

- 2/6 said they would not create a pact for small tasks due to effort required

Interpretation:

While the app aims to foster accountability, the high friction of the setup process violates the "Flexibility and Efficiency of Use" heuristic. This is a high severity issue, as it creates a significant barrier to entry at precisely the moment users are most likely to avoid the task.

Recommendation:

- Provide smart defaults and suggested options for task type, verification, and consequence.
- Offer a “quick pact” mode for small or low-stakes tasks to reduce activation energy.

Next Steps

1. Redesign Pact Creation Flow to optimise pact creation
    a. Allow new users to begin immediately with minimal decision-making required.
    b. Allow advanced users to customise their settings according to their preferences.
2. Improve consequence page by introducing “Random Consequence” feature
    a. Makes the pact feel more serious and engaging
3. Add additional features such as a coin shop.
4. Conduct user testing with 12 users with the revised prototype

Appendix

Validation Run

Participants

Participant 1 – Adrian, 24 (Completed task)

Void Deck: 19 Feb, 0900 - 0920 & 20 Feb 1000 - 1015

Observation Notes:

- Creation of pact took 55 seconds and submission of evidence took 25 seconds
- Found the pact creation screen easily
- Slightly frustrated and stated that pact creation was too “long”, and “not suitable for simple tasks”
- Chose the do push-ups consequence.

Interview Questions Day 1:

- When do you plan to start?
    - Tonight at 9pm
- Do you think the pact will affect your behavior towards doing the task?
    - Yes. It will keep reminding me to complete my task, so I can’t make an excuse that I forgot about it.

Interview Questions Day 2:

- What factors influenced when or whether you started the task?
    - The app, and that my task needs to be completed whether the app exists or not
- How would you describe your experience while completing the task?
    - It was alright, maybe I felt slightly more motivated.
- Do you think the pact affected your behavior? How or why?
    - Yes, I started an hour earlier than I thought I would
- What makes you pick this consequence?
    - It was simple and doable even if I failed to complete my task.

Comments

- The participant noted the app had an influence in him completing the task, but has an issue with the steps needed to complete the pact creation.

Participant 2 – Ken, 22 (Completed task)

Online: 20 Feb, 15:15 - 15:35 & 21 Feb 14:05 - 14:20

Observation Notes:

- Creation of pact took ~2 minutes and submission took ~30 seconds
- Chose a simple consequence of displaying “Sloth” badge on user profile.

Interview Questions Day 1:

- When do you plan to start?
    - Tomorrow morning around 9am
- Do you think the pact will affect your behavior towards doing the task?
    - Slightly, but since the task is simple I don’t think it would be play too big of a role

Interview Questions Day 2:

- What factors influenced when or whether you started the task?
    - The task was simple so I didn’t need much motivation, but the app did help me be a bit more engaged and reminded me to start it
- How would you describe your experience while completing the task?
    - Very straightforward and manageable,
- Do you think the pact affected your behavior? How or why?
    - The pact itself didn’t significantly affect my behavior, because the task didn’t take much effort but for bigger/longer tasks it would have
- What makes you pick this consequence?
    - The task wasn’t too complex so I chose a simple consequence to match it

Comments

- Participant noted that the pact creation process was relatively lengthy and may not be suitable for short or simple tasks

Participant 3 – Darren, 23 (Did not complete task)

Online: 20 Feb, 16:00 – 16:25 & 21 Feb, 15:30 – 15:45

Observation Notes:

- Pact creation took 2 minutes, participant visibly hesitated on multiple screens and remarked that the process felt long
- Chose a simple, low-stakes consequence (skipping one gym session)
- Appeared relaxed during Day 1; did not seem to view the consequence as a deterrent
- Did not submit evidence on Day 2 as task was not completed; walked through the flow as a simulation

Interview Questions Day 1:

- When do you plan to start?
    - tonight, around 11pm after I'm done with my other stuff.
- Do you think the pact will affect your behavior towards doing the task?
    - Maybe a little, but I'm not too worried. The consequence I picked isn't that bad so it's not like I'd be in trouble if I didn't finish.

Interview Questions Day 2:

- What factors influenced when or whether you started the task?
    - I kept pushing it back. I told myself I'd start at 11 but then got distracted and ended up not starting at all.
- Since the task was not completed, what barriers prevented completion?
    - Honestly, I underestimated how tired I'd be. And because the consequence wasn't something I really cared about, there wasn't enough pressure to push through.
- Do you think the pact affected your behavior? How or why?
    - Not really. I was aware of the pact but it didn't create enough urgency. If the consequence had been something more embarrassing or involved my friends, I think it would have had more of an effect.
- What makes you pick this consequence?
    - I just chose something quick and easy. I didn't want to pick something too harsh in case I actually failed, which I did. I probably should have picked something that actually scared me a bit.

Comments

- Participant noted that the pact creation process felt long, particularly for a task he considered relatively minor. He suggested that a faster, simplified flow would be more practical for everyday use. He also mentioned that his choice of a simple, low-impact consequence may have reduced the pact's effectiveness, and acknowledged that a more meaningful or social consequence would likely have motivated him more.

Participant 4 – Felix, 21 (Did not complete task)

Online: 20 Feb, 17:00 – 17:20 & 21 Feb, 16:00 – 16:15

Observation Notes:

- Pact creation took 2 minutes
- Chose a simple, low-stakes consequence (no social media for one evening)
- Seemed fairly relaxed throughout Day 1, and she did not appear particularly concerned about the consequence
- Did not submit evidence on Day 2 as task was not completed; walked through the flow as a simulation

Interview Questions Day 1:

- When do you plan to start?
    - After dinner, around 8 pm
- Do you think the pact will affect your behavior towards doing the task?
    - Yes, I do think it will affect as I made the pact with someone else

Interview Questions Day 2:

- What factors influenced when or whether you started the task?
    - I kept delaying it after dinner. I told myself I'd start soon but one thing led to another and before I knew it, it was too late.
- Since the task was not completed, what barriers prevented completion?
    - I think the main issue was that the consequence didn't feel serious enough to push me. I knew in the back of my mind that even if I failed, it wasn't a big deal, so I didn't feel the urgency to start.
- Do you think the pact affected your behavior? How or why?
    - Not really. I was aware of it but the consequence I chose was too easy to accept. If I had picked something that actually mattered to me like something my friends would see, I think I would have been more motivated.
- What makes you pick this consequence?
    - I didn't want to commit to anything too uncomfortable just in case I didn't manage to finish. So I played it safe, which probably worked against me.

Comments

- Participant reflected that her choice of a simple, low-impact consequence significantly reduced the pact's effectiveness. She acknowledged that she deliberately avoided picking a meaningful consequence as a safety net, which ultimately removed the pressure needed to follow through. She suggested that having fewer but more impactful consequence options might prevent users from defaulting to easy choices.

Participant 5 – Marcus, 22 (Completed task)

Online: 20 Feb, 18:00 – 18:30 & 21 Feb, 17:00 – 17:15

Observation Notes:

- Pact creation took 1 minute
- Chose a simple, low-stakes consequence (skipping his afternoon coffee for a day)
- Completed the task and submitted evidence on Day 2 without issues
- Stated unprompted that he finished the task out of habit and personal discipline rather than because of the pact

Interview Questions Day 1:

- When do you plan to start?
    - Around 9 pm. Usually get things done when I sit down to work.
- Do you think the pact will affect your behavior towards doing the task?
    - Honestly, probably not that much. I would have done the task anyway. But it's interesting to have something tracking it.

Interview Questions Day 2:

- What factors influenced when or whether you started the task?
    - Just my usual routine. I sat down at my desk after dinner and got into it. I don't really need much external motivation for tasks like this.
- How would you describe your experience while completing the task?
    - It went smoothly. I finished within an hour, which is about what I expected.
- Do you think the pact affected your behavior? How or why?
    - Not really. I completed the task but it wasn't because of the pact or the consequence. I would have done it regardless. The consequence I chose wasn't something I was worried about losing.
- What makes you pick this consequence?
    - The task wasn't something I was dreading so I just picked whatever was quick and easy. I didn't feel the need to set a harsh consequence for myself.

Comments

- Participant noted that the pact creation process felt unnecessarily long, especially for a straightforward task. He also pointed out that the consequence played no role in his decision to complete the task, his completion was driven entirely by personal habit and routine, highlighting that the current consequence mechanism may not add value for self-motivated users.

Participant 6 – Orvin, 19 (Completed task)
SMU Campus: 19 Feb, 1500-1515 – 21 Feb 1115-1125

Observation Notes:
- The participant chose a simple consequence: he was able to use the app smoothly without hindrance. (has plenty of experience using a variety of mobile apps)
- Chose push-ups as his consequence.
Interview Questions Day 1:
- When do you plan to start?
  - Probably tonight
- Do you think the pact will affect your behavior towards doing the task?
  - Maybe, I guess it adds a certain amount of playful pressure to make sure i finish it
Interview Questions Day 2:
- What factors influenced when or whether you started the task?
  - I think the threat of the task itself was the biggest reason, not really the consequence. I think the consequence was just an added “fun” bonus to completing my duties
- How would you describe your experience while completing the task?
  - I think that the looming consequence was always at the back of my mind whilst I was doing it. It was both funny and useful in a certain way. It definitely stopped me from procrastinating a couple of times throughout.
- Do you think the pact affected your behavior? How or why?
  - I think it did, for some reason or another I was more inclined to fully finish my task in one go instead of procrastinating like usual and finishing it in short bursts throughout the day.
- What makes you pick this consequence?
  - It was something that was simple and wouldn’t be detrimental if I didn’t accomplish it, but at the same time it wouldn’t be the best idea to just let it happen.
Comments
- The participant mentioned that it’s not worth going through the hassle of creating a pact and setting up consequences/accountability partners for tasks that takes only a couple of hrs/mins to finish.

Validation Materials

Informed consent

“Thank you for participating in our study.

We’re testing a prototype app designed to help reduce procrastination through accountability pacts.

Today’s session will take about 20 minutes, and we’ll have another 20-minute session tomorrow. You’ll create a pact for a real task you’re currently avoiding, and tomorrow you’ll try to submit evidence for the completed pact.

Your participation is completely voluntary. You may stop at any time or skip any question.

We may record this session for analysis purposes, but the recording will not be publicly released. It will only be used for academic evaluation. Your name, age, and gender may be included in the report.

You may feel mild pressure from committing to a task, but there are no major risks involved.

Do you understand and agree to participate?”

Task instructions

Pre-condition:
Participants were recruited on the condition that they already had a task due within 48 hours that they intended to complete.

Day 1:

Starting from the homepage:

1. Navigate to the pact creation screen and create a pact for your identified task.

2. Set the deadline as today.

3. Select Human verification as the verifying method.

4. Choose a friend to hold you accountable (for testing purposes, select “Jun Wei”).

5. Select any consequence you feel is appropriate.

Day 2:

Starting from the homepage:

1. Locate the pact you created.

2. If you completed the task, submit evidence using the prototype.

3. If you did not complete the task, simulate submitting evidence.

Interview Questions

Day 1:
1. When do you plan to start?
2. Do you think the pact will affect your behavior towards doing the task?

Day 2:
1. What factors influenced when or whether you started the task?
2. (If a participant completed his task) How would you describe your experience while completing the task?
3. Do you think the pact affected your behavior? How or why?
4. (If a participant did not complete his task) Since the task was not completed, what barriers prevented completion?
5. What makes you pick this consequence?

Others

Participants will complete their tasks on a laptop, using the Figma prototype.

The session will be recorded for analysis purposes only.

Participants are encouraged to think aloud and verbalize their thoughts during the session.

Data Analysis

Summary of the analysis

The user testing (n = 6) indicates that while the Pact concept demonstrates moderate behavioural influence, its effectiveness is constrained by two primary issues:

- High setup friction during pact creation.

- User self-selection of low-stakes consequences, which weakens accountability.

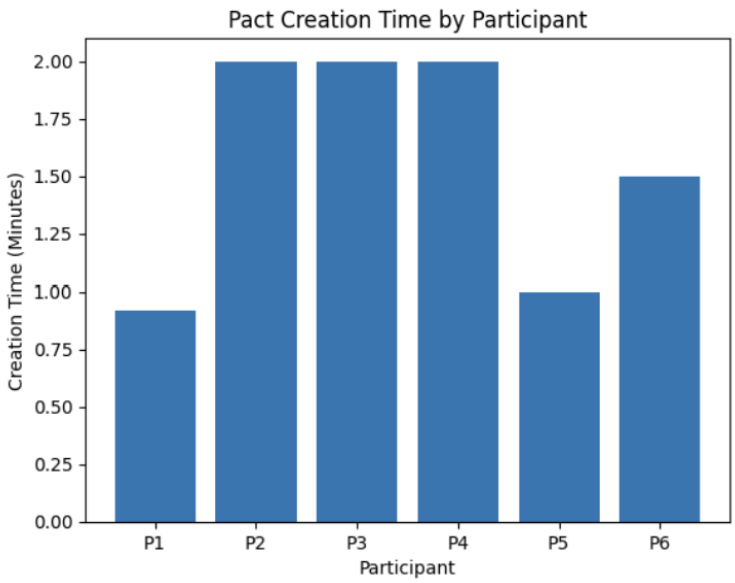
Although the overall task completion rate was 66.7% (4 out of 6 participants), notably, half of those who completed their tasks reported that the pact meaningfully influenced their behaviour.

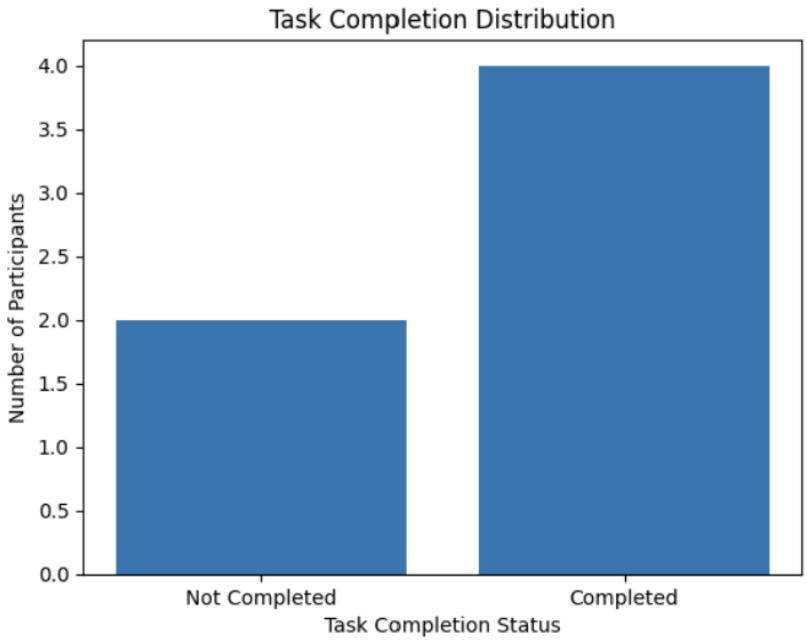
This indicates that the system has demonstrated the ability to positively impact user behaviour, particularly among participants who are moderately prone to procrastination. The findings

suggest that when users perceive sufficient accountability, the pact can serve as an effective behavioural nudge.

While not all completions were directly driven by the pact, the results validate the concept's potential and highlight opportunities to strengthen its accountability mechanisms for broader impact.

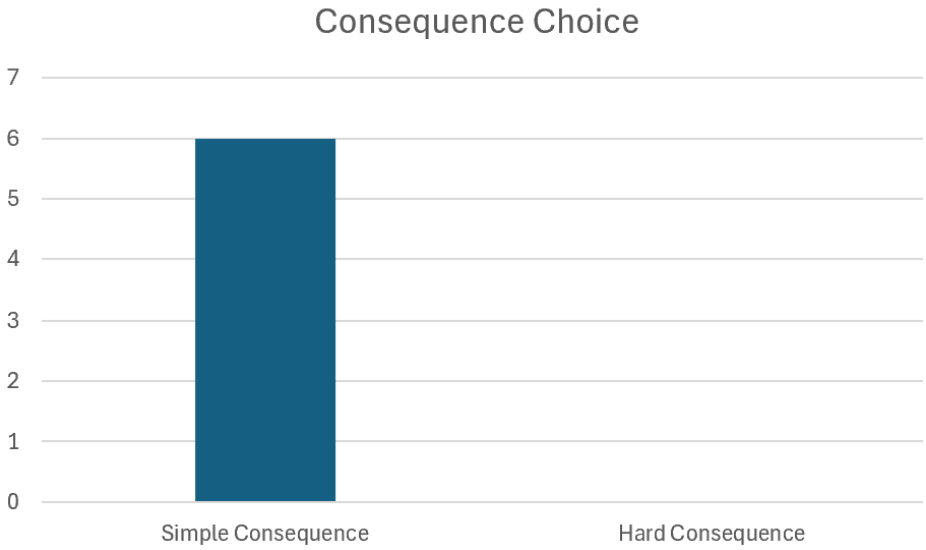
Analysis

<table>
<tr><td>Pact Creation Time</td><td> 
</td></tr>
<tr><td colspan="2">Average creation time: 1.57 minutes
Despite being under 2 minutes, the majority of participants described the process as:
- "Long"
- "Lengthy"
- "Not suitable for simple tasks"
- "Unnecessarily heavy"</td></tr>
</table>

Task Completion Rate	

- Completed: 4/6 (66.7%)
- Not completed: 2/6 (33.3%)

The 2 participants who failed mentioned weak consequences reduced urgency.

Consequence Severity Pattern	

All 6 participants selected low-stakes consequences

- 2 participants chose exercise

- 2 chose to skip an activity
- 1 went on an evening social media detox
- 1 chose to let the app display “Sloth” badge

Common reasoning:
- “Didn’t want something too harsh.”
- “Played it safe.”
- “Not too serious in case I failed.”

Recommendation

- Externalize Consequence Selection

All 6 participants selected low-severity consequences ("Mercy selection"), admitting they chose things they "wouldn't mind doing." This rendered the accountability mechanism ineffective.

Recommendation: Implement "Mystery" or "Random" Consequences. Allow users to commit to a severity level (e.g., low, medium and high) without knowing the specific penalty until failure.

Why would this be helpful: This restores the "psychological cost" of failure. By removing user control over the specific penalty, we eliminate the internal bargaining that undermines behavioral change.

- Streamline the "Quick-Pact" Architecture

The current 10+ step flow is a high severity issue. Many users felt the process was too long, creating a high barrier to entry that mirrors the effort of the task itself.

Recommendation:

Fast Pact Creation: Develop a simplified mode using Smart Defaults (e.g., default to "Selfie" verification and "10-minute" reminders).

Progressive Disclosure: Hide advanced customization (human verification, detailed scheduling) behind an "Advanced" toggle for power users.

Why would this be helpful: According to the Efficiency of Use heuristic, reducing the "activation energy" ensures that the app is easier to use than the task is to avoid. This prevents the app itself from becoming a target for procrastination.